Linear Regression (Continued)

In Linear Regression → Training a model
↓
creating a best-fitting line

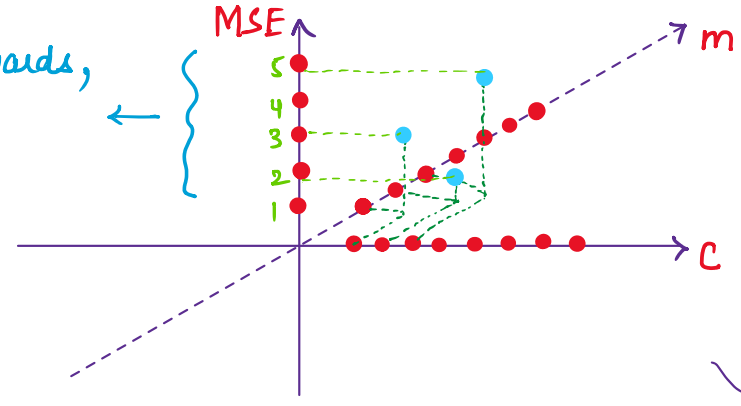
- Why best-fit line?
  ↳ On best-fit line, we get predictions

- How to create best-fit line?
  ↳ By using best values of m & c

Let us suppose for one m, & c, value we get MSE,
Similarly, for multiple m & c we get multiple MSE

so, $m_1, c_1 \longrightarrow MSE_1$
$m_2, c_2 \longrightarrow MSE_2$
$m_3, c_3 \longrightarrow MSE_3$
⋮ ⋮ ⋮

↳ let us draw a graph b/w m, c & MSE



On moving upwards, MSE increases
But we want low MSE
↓
(moving in downward direction)

MSE is dependent on m (slope) and c (intercept)

downward direction)

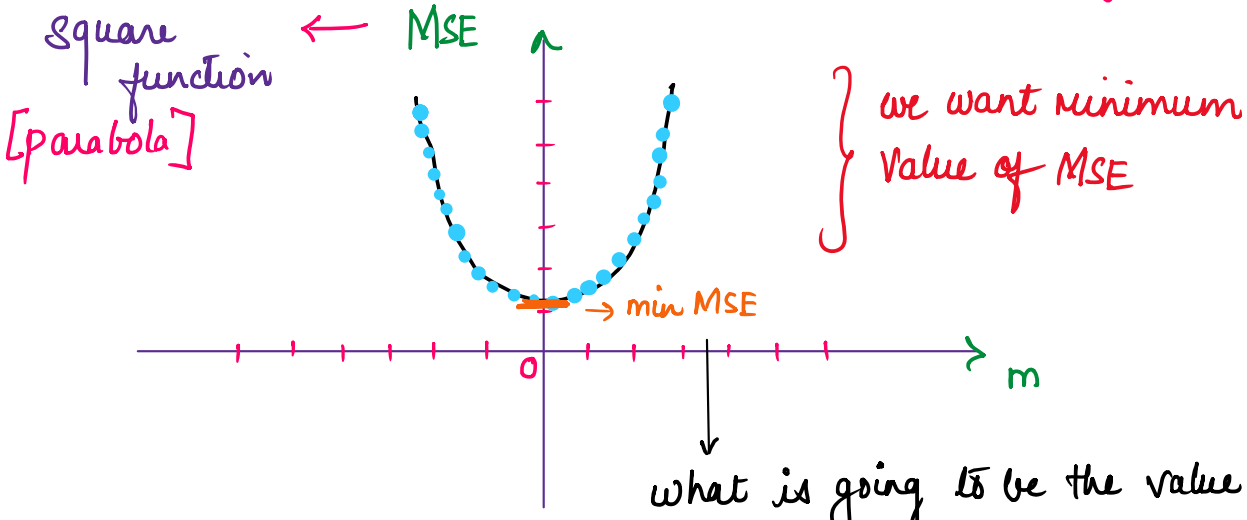
Using above logic we can say,
we want that particular value of slope (m) & intercept (c) where we get minimum (lowest) MSE

Above graph is 3D
↳ To make it simple, we use 2D

Let us suppose, we take c = 0

so, now graph will have 2 axis
- m (independent)
- MSE (dependent)

easier to understand using 2D



what is going to be the value of m here

↓

for this we will use an algorithm that is known as "Gradient Descent Algorithm"